7.2 Activities: column methods and long division

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Written methods can feel dry. These activities give them a reason and a story. Always use the layout he uses at school.

1. Receipt detective

You need: old supermarket receipts (*kassabonnen*), a pencil.

How: Cover the total with a sticky note. He picks three or four items and adds them in columns. Then he estimates the full total by rounding each price to whole euros, and finally checks against the real total. Did the shop get it right?

Talk about it: "How close was your estimate? Which item made the biggest difference?"

2. Shop till

You need: a few items from the cupboard with price stickers (write prices like € 2,49 or € 0,85), play money or real coins.

How: He is the cashier. You "buy" two or three items and pay with a € 10 or € 20 note. He adds the prices in columns and works out the change by subtracting (*komma onder komma*), then counts out the change with coins to check. Swap roles: you make a (deliberate) mistake now and then for him to catch.

Talk about it: "How did you know the comma goes there?" "Is there a quicker way to find the change?" (Counting up from the total is great.)

3. Area model on squared paper

You need: squared paper, coloured pencils.

How: Draw a rectangle 23 squares by 46 squares (or a smaller one, like 13 by 24, to start). Split it with lines into 20 + 3 and 40 + 6. Colour the four parts and count each part using times tables (20 × 40, 20 × 6, 3 × 40, 3 × 6). Add the parts. Then do the same sum the school way and see where the numbers on the lines of the written method come from.

Talk about it: "Which part of the rectangle is the 3 × 46 line? Which is the 20 × 46 line?"

4. Sharing the treasure

You need: play money in € 100, € 10 and € 1 notes/coins (or paper slips), 3–6 cups as "pirates".

How: Make a treasure of, say, € 672 using 6 hundreds, 7 tens and 2 ones. Share it fairly between 6 pirates. Start with the hundreds. When the tens don't share evenly, swap a ten for ten ones (*inwisselen*). Write down each step next to the school's layout. Start with happy numbers that share exactly: € 848 between 4, € 936 between 3.

Talk about it: "When did you have to swap? Where can you see that in the written sum?"

5. What do we do with the rest?

You need: a bag of 50 or so small things (dried beans, Lego, sweets), egg boxes or cups.

How: Make up little stories and act them out. "We have 50 beans and boxes of 6. How many boxes can we fill?" (8, with 2 left over.) "We need to pack them all. How many boxes do we need?"

(9.) “Share 50 euro cents between 4 people.” (12,5 cents: then talk about why you can’t really do half a cent.)

Talk about it: “Same sum, different answer. Why?” This is exactly what groep 7 word problems test.

6. Make it true

Play “Make it true” from GAMES.md with column sums: write a column sum with some digits missing and take turns filling them in.

Conversation prompts for any day

- “Before you start, roughly what will the answer be?”
- “How could you check this answer in a different way?” (Multiply back after dividing; add back after subtracting.)
- “Which way of dividing do you like best, big bites or small bites? Why?”